Topology-Aware Distributed Layerwise Offloading for MiniMax-H3 on Eight NVIDIA B300 GPUs

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Abstract

We study how data parallelism (DP), sequence parallelism (SP), and distributed layerwise offloading (DLO) interact for MiniMax-H3 video–audio generation on one eight-GPU NVIDIA B300 node. Three production-relevant routes are evaluated with 20 measured 50-step text-to-video-and-audio (T2VA) waves across two engine lifecycles. DP1×SP8 with AllGather minimizes wave latency (P50/P95: 34.55/35.25 s), DP4×SP2 with AllGather is the balanced point (151.89 videos/h at P95 95.31 s), and DP8×SP1 with rank-local DLO maximizes throughput and minimizes energy (183.78 videos/h, 43.97 Wh/video). The central result is that DLO mode is topology-dependent: AllGather improves DP1 throughput by 129.4%, offers only 2.2% at DP4, and reduces DP8 throughput by 4.1% while increasing the observed per-GPU memory peak from 20.03 to 94.03 GiB. FL2VA and Ref2VA preserve the same latency–throughput ordering. These measurements reject a single global DLO policy and motivate topology-aware deployment.

Keywords: diffusion inference; distributed layerwise offload; data parallelism; sequence parallelism; video generation; B300; vLLM-Omni

1 Introduction

On a fixed multi-GPU node, sequence parallelism reduces one request's latency, whereas data parallelism increases concurrent request capacity. Distributed layerwise offloading adds a second design axis: DP replicas may advance weights through compatible collective waves (AllGather) or maintain rank-local weight state and execute independent requests. Communication, host-to-device transfer, activation memory, and concurrency therefore change together.

This note asks three questions: **RQ1**: Which evaluated DP×SP configurations lie on the latency–throughput frontier? **RQ2**: At which topology does AllGather outperform rank-local DLO? **RQ3**: Does the frontier persist under image- and audio-conditioned generation? vLLM-Omni supports diffusion and heterogeneous multimodal outputs [1]; recent changes harden collective DP waves [4] and enable independent rank-local DP requests [5].

Contributions. We provide (i) a full-step, two-lifecycle selected-route T2VA frontier with latency, throughput, energy, and memory; (ii) a paired AllGather/rank-local comparison; and (iii) FL2VA/Ref2VA transfer measurements with preserved wave-level data.

2 Method

The long-run protocol evaluates DP1×SP8, DP4×SP2, and DP8×SP1. DP2×SP4 was omitted when the requested long-run protocol was narrowed to three recommended routes; it was not measured and is not claimed to be dominated. Consequently, frontier statements are conditional on the evaluated set. DP1, DP4, and DP8 generate one, four, and eight outputs per wave. Throughput is measured outputs divided by measured wave time. Per-video energy is the trapezoidal integral of summed eight-GPU board power divided by output count; no idle baseline is subtracted. We report empirical P50/P95 and CV. The $n = 20$ mean-latency t-intervals are descriptive because samples are clustered within two engine lifecycles, not independent production trials. Output video and audio shapes are validated for every measured request.

Table 1: Controlled experimental configuration.

Dimension	Setting
Node	8× NVIDIA B300 SXM6 AC; NV18 full mesh; one NUMA domain
Model	MiniMax-H3 FL2VA / Ref2VA; BF16; `CUDNN_ATTN`
Output	768×1344; 124 frames; stereo audio `[1,2,165600]`
DLO	`resident_layers=0`; AllGather or rank-local; eager execution
Sampling	50 requested steps (49 scheduler denoising updates)
T2VA	1 full warmup/lifecycle; 5+15 measured waves; 2 lifecycles
FL2VA / Ref2VA	1 full warmup; 5 measured waves per selected route
Telemetry	External `nvidia-smi`; 0.758 s median interval

Figure 1 summarizes the execution dynamics described by the DLO design [2]. DP and SP form a two-dimensional process grid: SP exchanges activations within a request, whereas DLO AllGather reconstructs the next block across the DP group (or the SP group when DP=1). With `resident_layers=0`, two device buffers alternate while block N computes and block $N+1$ is prefetched. Rank-local DLO removes this weight collective and permits replicas to progress independently, but retains full rank-local host tensors.

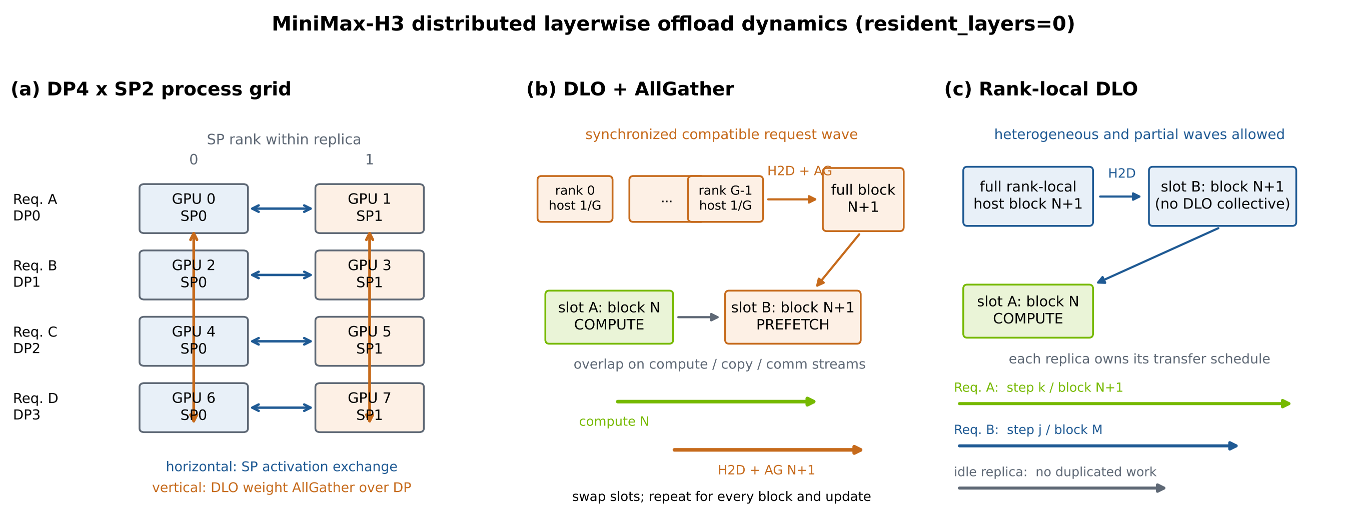

Figure 1: MiniMax-H3 DLO topology and per-block dynamics. The DP4×SP2 example exposes the orthogonal SP activation and DLO weight-collective groups; the two right panels contrast synchronized AllGather waves with independent rank-local execution.

The tested source is 9e73ee1 plus a recorded local subgroup-broadcast fix. The runtime used PyTorch `2.11.0+cu130`; it warned that vLLM-Omni `0.26.0rc2.dev11` and vLLM `0.24.0` were not release-aligned.

3 Results

3.1 RQ1: Selected-route latency–throughput frontier

Table 2: Selected T2VA routes ($n = 20$ measured 50-step waves per row).

Route	P50 (s)	P95	Videos/h	CV (%)	Peak/GPU (GiB)	Wh/video
DP1×SP8 / AG	**34.55**	**35.25**	103.84	1.38	26.37	68.08
DP4×SP2 / AG	94.73	95.31	151.89	0.35	25.11	51.76
DP8×SP1 / RL	156.74	157.03	**183.78**	**0.15**	**20.05**	**43.97**

All three measured routes are Pareto-optimal within the evaluated set (Fig. 2). DP2×SP4 remains an unmeasured candidate and could change the hull or the location of the knee. Relative to DP1, DP4 gains 46.3% throughput; relative to DP4, DP8 gains another 21.0% but raises P50 latency by 65.5%. Descriptive 95% t-intervals for mean wave latency are [34.440, 34.899] s, [94.646, 94.960] s, and [156.594, 156.824] s. Thus DP4 is the knee, not a universal optimum: DP1 serves low-latency traffic, whereas DP8 requires a continuously populated queue to realize its node-throughput advantage.

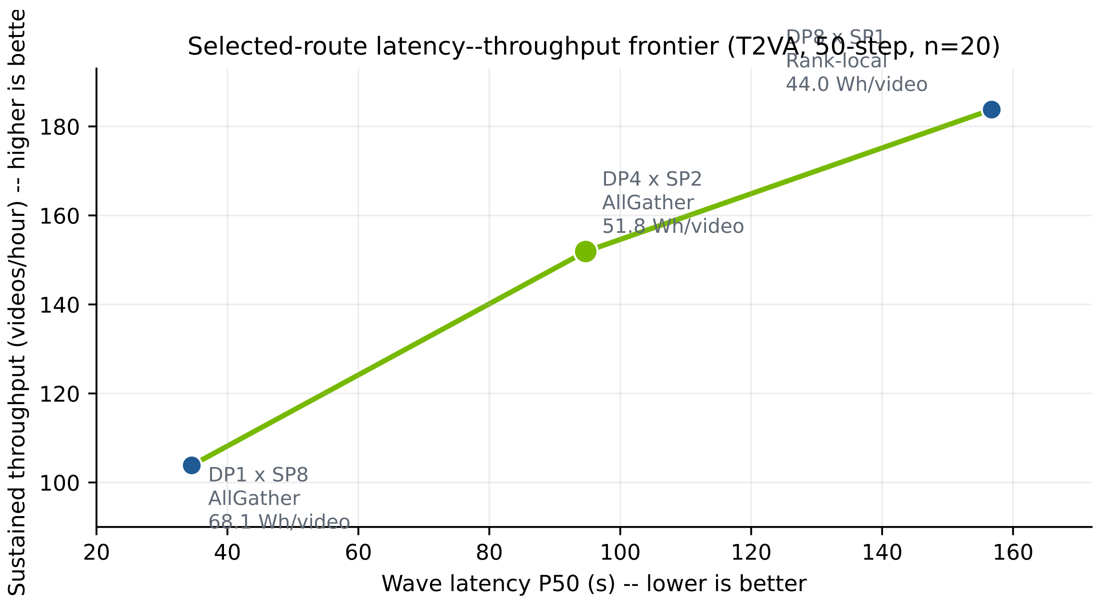

Figure 2: T2VA latency–throughput frontier. Labels include measured eight-GPU board energy per output.

3.2 RQ2: DLO mode is topology-dependent

Table 3: Effect of enabling AllGather relative to rank-local DLO ($n = 5$ paired waves/mode). Negative latency is favorable.

Topology	Throughput	P50 latency	Peak/GPU (RL→AG)	Decision
DP1×SP8	**+129.4%**	**–56.6%**	26.26→23.53 GiB	AllGather
DP4×SP2	+2.2%	–2.2%	23.91→24.97 GiB	AllGather
DP8×SP1	–4.1%	+3.8%	20.03→94.03 GiB	Rank-local

At DP1, AllGather uses the SP group and is not a no-op. It simultaneously reduces latency, energy (28.0%), and measured peak memory. At DP4 its gain is small. At DP8 it is strictly dominated in this workload and produces a large asymmetric memory peak (Fig. 3). The data therefore reject topology-independent DLO policies.

3.3 RQ3: Transfer to conditioned generation

Table 4: Best route by service objective for each workload ($n = 5$/route).

Workload	Lowest P50 latency	Highest throughput	Lowest energy
FL2VA first-frame	DP1/AG: 37.92 s	DP8/RL: 165.93 videos/h	DP8/RL: 46.88 Wh
Ref2VA image+audio	DP1/AG: 55.44 s	DP8/RL: 100.07 videos/h	DP8/RL: 74.06 Wh

Conditioning changes scale but not direction: DP1/AllGather remains the latency route and DP8/rank-local remains the throughput/energy route (Fig. 4). Memory does not transfer as cleanly. Ref2VA AllGather approaches 100 GiB/GPU, while DP8/rank-local measures 19.36 GiB.

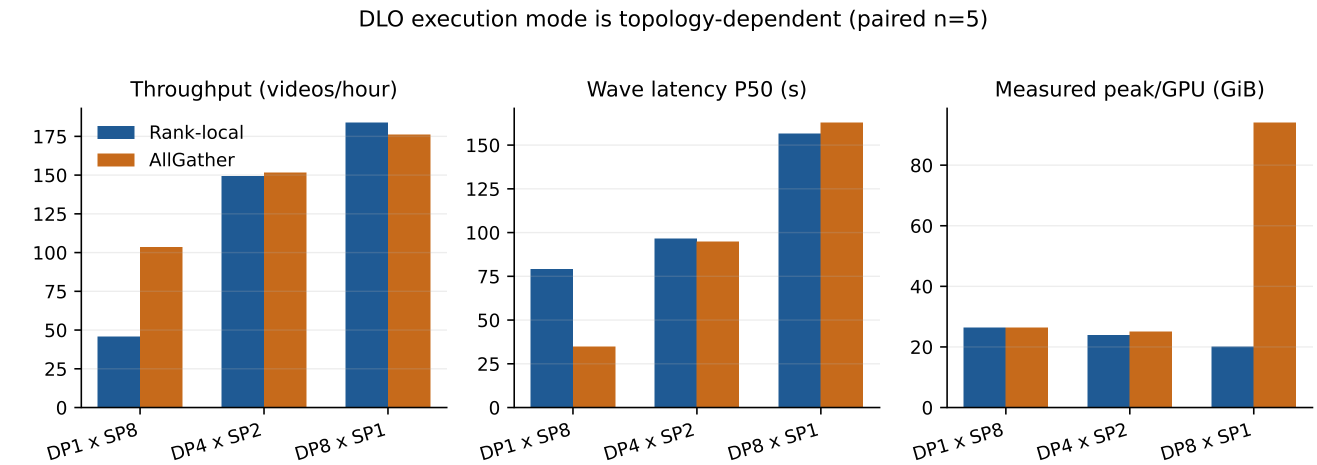

Figure 3: Paired DLO-mode comparison. AllGather's benefit reverses at DP8.

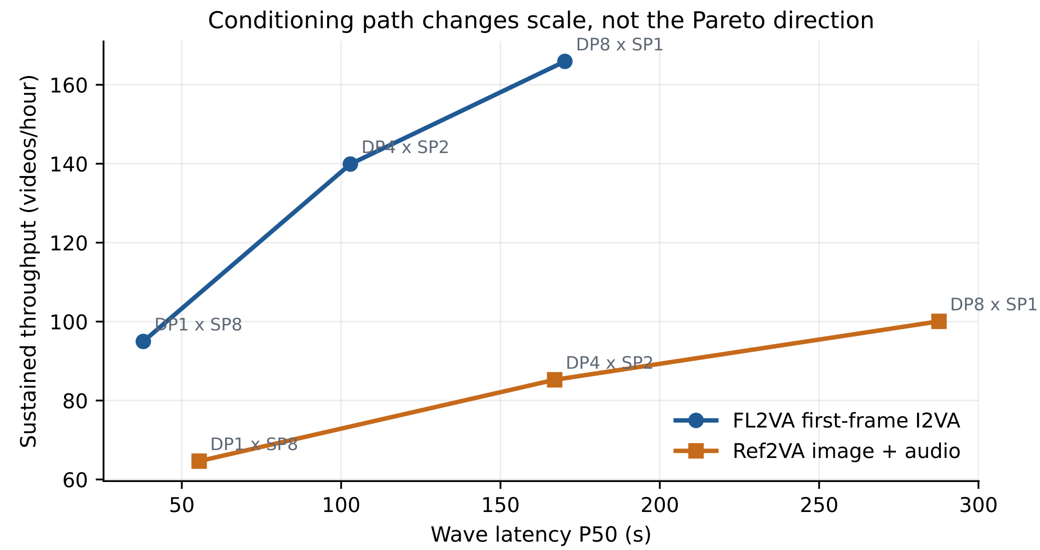

Figure 4: FL2VA and Ref2VA frontiers preserve the T2VA ordering.

4 Discussion

The systems interpretation is simple. SP converts eight GPUs into latency reduction for one request; DP converts them into concurrent service capacity. AllGather is useful when coordinated weight movement offsets transfer overhead, but becomes harmful when collective state and memory concentration dominate. Because this crossover occurs within one node and model, DLO mode should be selected as part of the topology, not as a model-wide switch.

The production policy supported by these measurements is: **DP1×SP8/AllGather** for latency-sensitive traffic, **DP4×SP2/AllGather** as the balanced default, and **DP8×SP1/rank-local** for sustained high queue depth.

5 Evaluation Integrity and Limitations

The first DP4×SP2 Ref2VA warmup exposed a correctness defect: subgroup collectives used global `src=0` when the group did not contain global rank zero. Eight broadcast sites were repaired locally using `dist.get_global_rank(group, 0)`. A two-step image+audio smoke passed before the complete five-wave rerun; failure, smoke, and rerun artifacts were kept separate. This fix is not claimed to be part of PR #5911.

All successful cases passed output-shape checks without OOM, worker death, segfault, attention operation-creation failure, or post-fix group-rank error. Seven cases exhibited a non-fatal 30 s orchestrator-stop delay after output; the final external check found all GPUs at zero memory and utilization.

The study is limited to three of the four natural eight-GPU DP×SP factorizations: DP2×SP4 is unmeasured, so the reported Pareto frontier is not a complete topology sweep. It also covers one B300 node, one input set, BF16, one resolution and frame count, and batch size one per replica. Board energy excludes host and facility power. Shape checks do not establish perceptual quality. Twenty T2VA waves across two lifecycles characterize steady state but do not constitute a long-duration SLA or failure-injection campaign. The local fix and version mismatch should be resolved before production qualification.

6 Artifact Availability

The HTML edition, this PDF, 15-case CSV, 105 wave samples, environment/input hashes, local source diff, and benchmark runners are archived at `https://github.com/lishunyang12/vllm-omni-rankings/tree/main/scripts/minimax_h3_b300_dlo_industrial_report`.

References

[1] P. Yin et al., *vLLM-Omni: Fully Disaggregated Serving for Any-to-Any Multimodal Models*, arXiv:2602.02204, 2026. `https://github.com/vllm-project/vllm-omni`.

[2] vLLM-Omni contributors, *Distributed Layerwise Offload*, design note, 2026. `https://github.com/vllm-project/vllm-omni/blob/main/docs/design/feature/distributed_layerwise_offload.md`.

[3] MiniMaxAI, *MiniMax-H3 model card*, 2026. `https://huggingface.co/MiniMaxAI/MiniMax-H3`.

[4] S. Li, *Fix DLO DP concurrent request execution*, vLLM-Omni PR #5864, 2026. `https://github.com/vllm-project/vllm-omni/pull/5864`.

[5] S. Li, *Enable independent requests for rank-local DLO DP*, vLLM-Omni PR #5911, 2026. `https://github.com/vllm-project/vllm-omni/pull/5911`.